INCH-POUND
MIL-DTL-24368/1C(NAVY)
27 July 2021
SUPERSEDING
MIL-C-24368/1B(NAVY)
25 March 1987

DETAIL SPECIFICATION SHEET

CONNECTOR ASSEMBLIES; PLUG, POWER TRANSFER, SHORE TO SHIP AND SHIP TO SHIP, 500 VOLTS, 500 AMPERES, 60 HERTZ

This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-C-24368.

All dimensions are in inches.

For tolerances other than those specified, see the dimensions and tolerances of MIL-C-24368.

AMSC N/A FSC 6150

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:33Z
Check the source to verify that this is the current version before use.

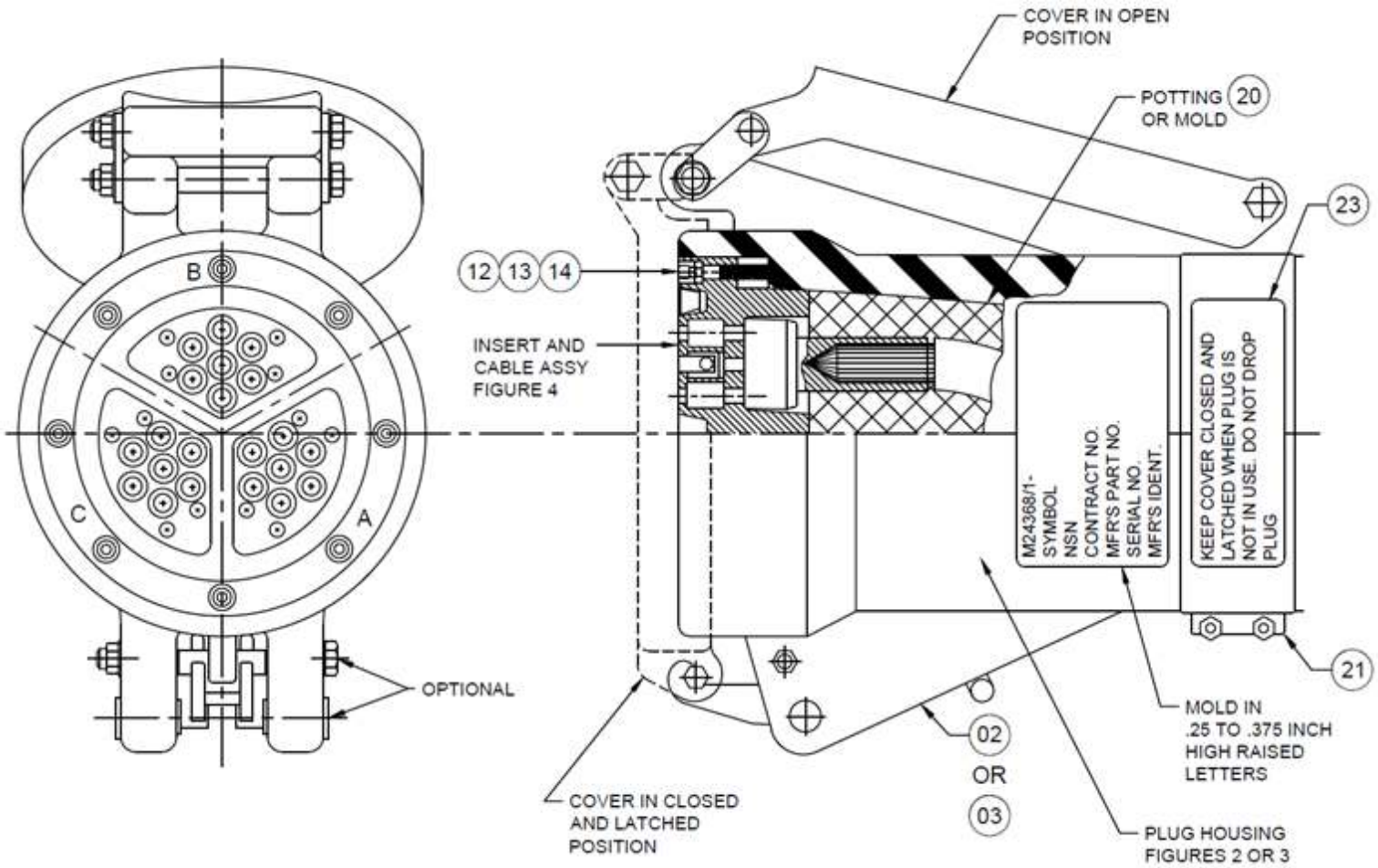


FIGURE 1. Plug, typical.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:33Z
Check the source to verify that this is the current version before use.

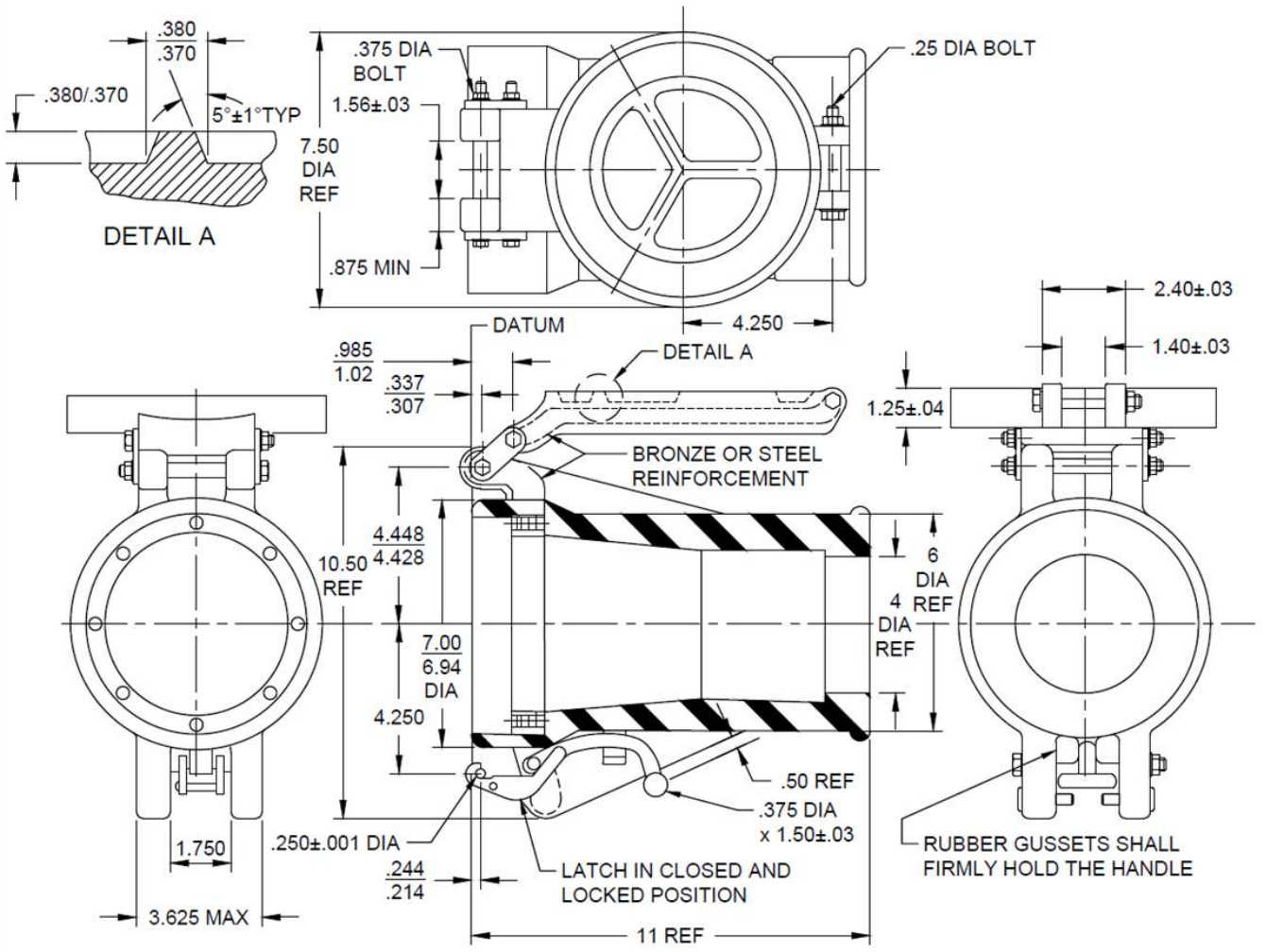


FIGURE 2. Plug housing assembly with cover.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:33Z
Check the source to verify that this is the current version before use.

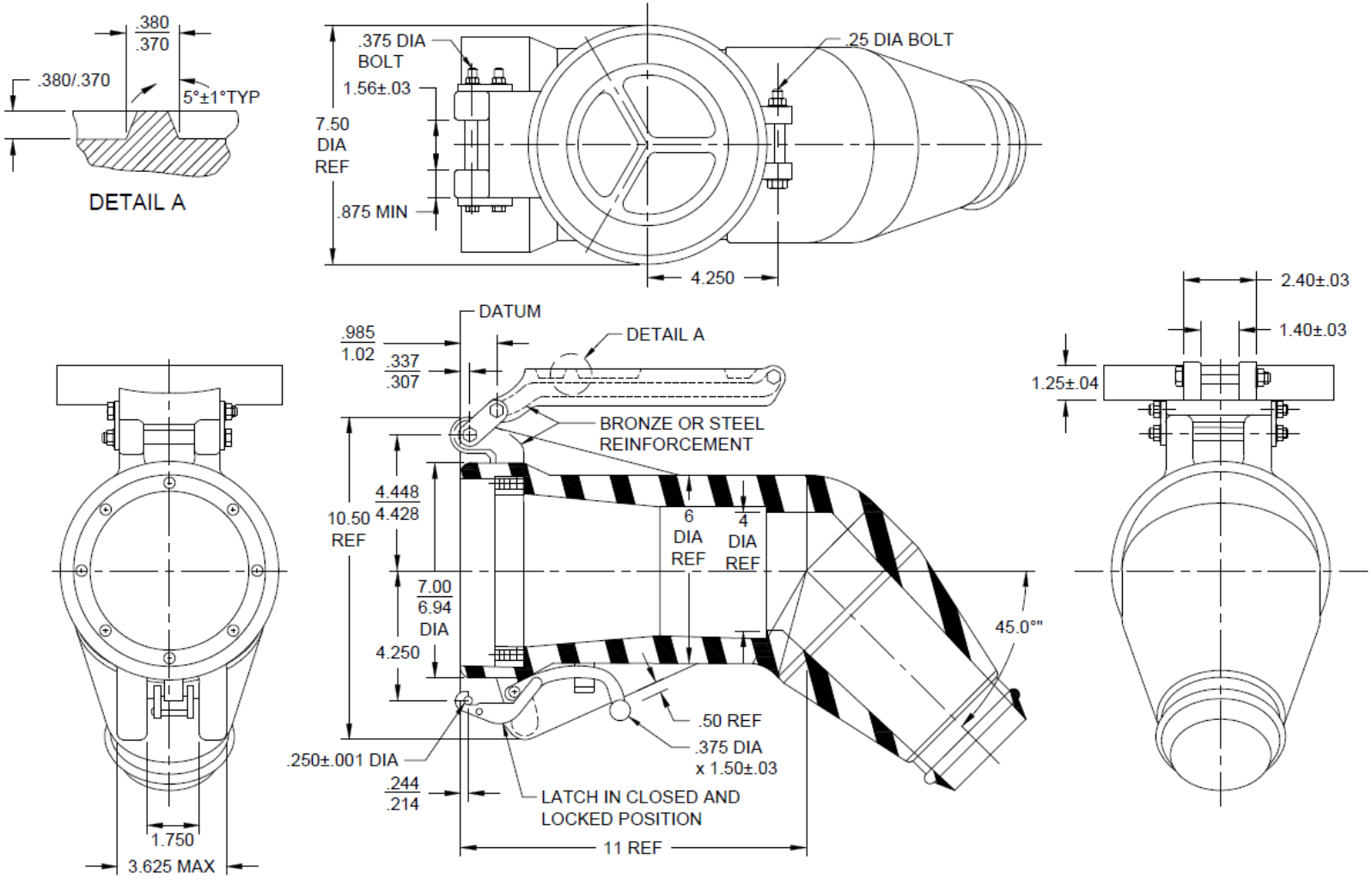
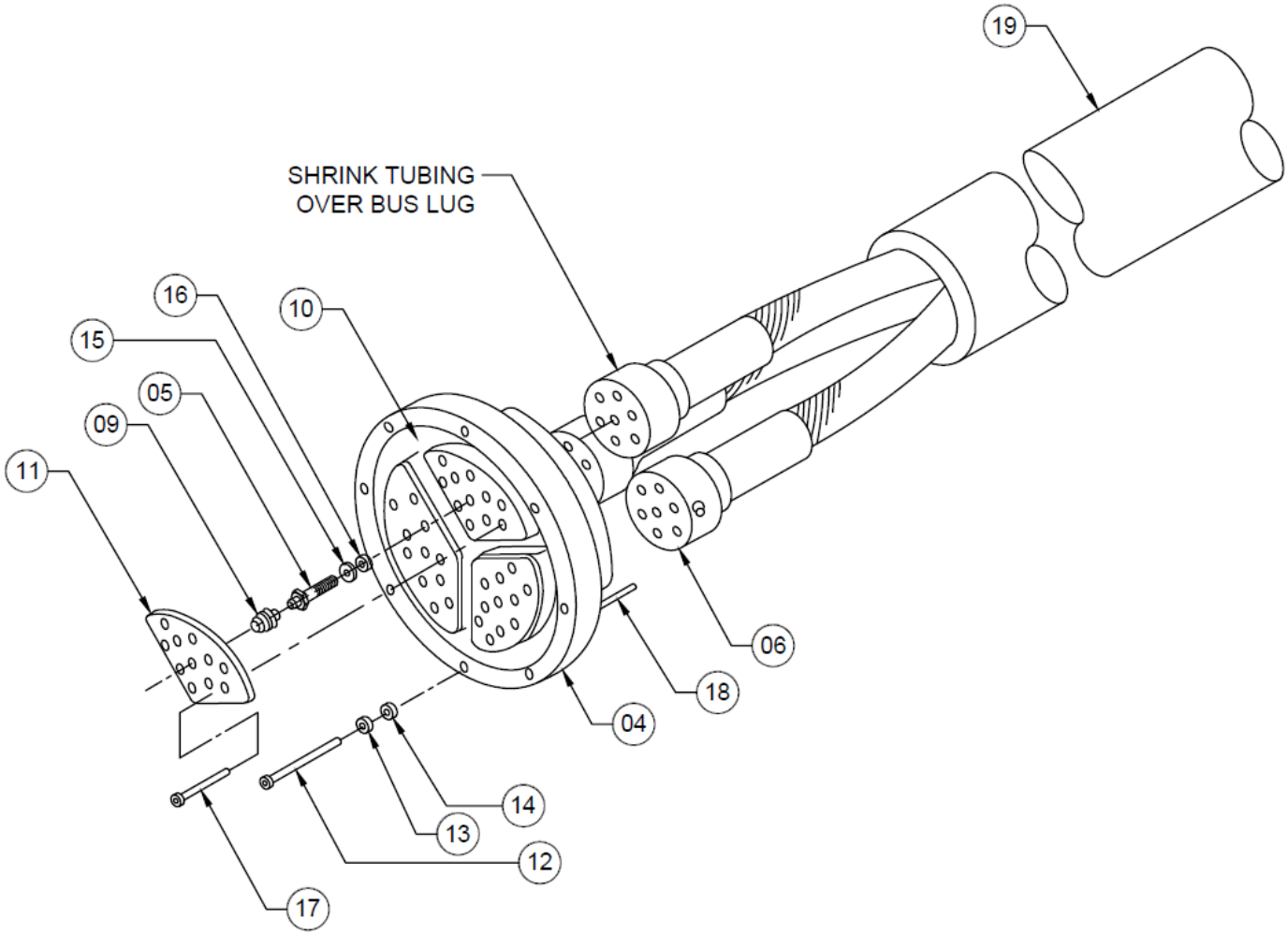


FIGURE 3. 45-degree plug housing assembly with cover (similar to figure 2).

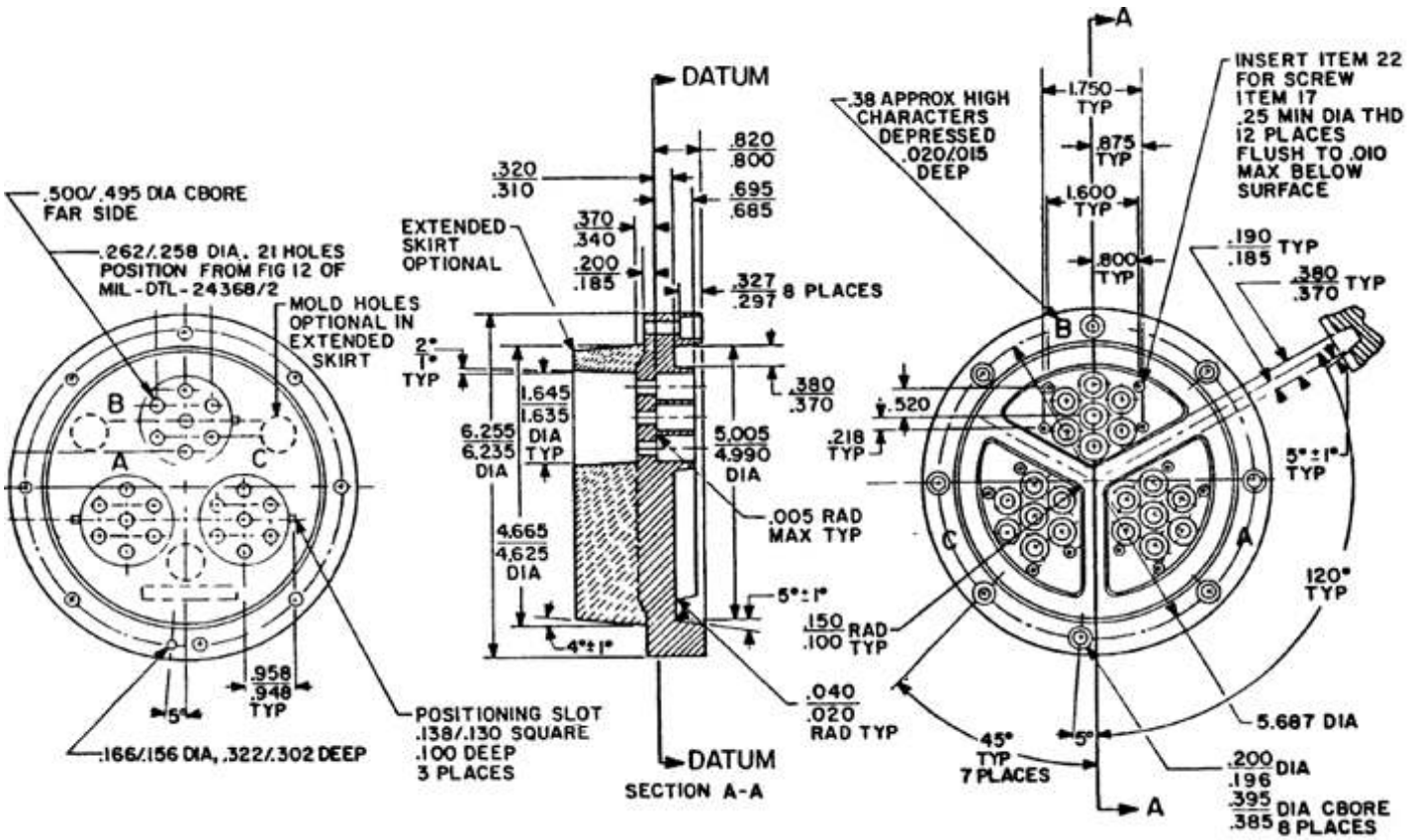
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:33Z
Check the source to verify that this is the current version before use.



NOTE: Cable termination procedures shall be in accordance with MIL-STD-2003-1. The minimum cable bend radius, as specified by MIL-HDBK-299, shall not be exceeded. Potting shall be in accordance with potting requirements herein (see requirement 6).

FIGURE 4. Insert and cable assembly, typical.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:33Z
Check the source to verify that this is the current version before use.

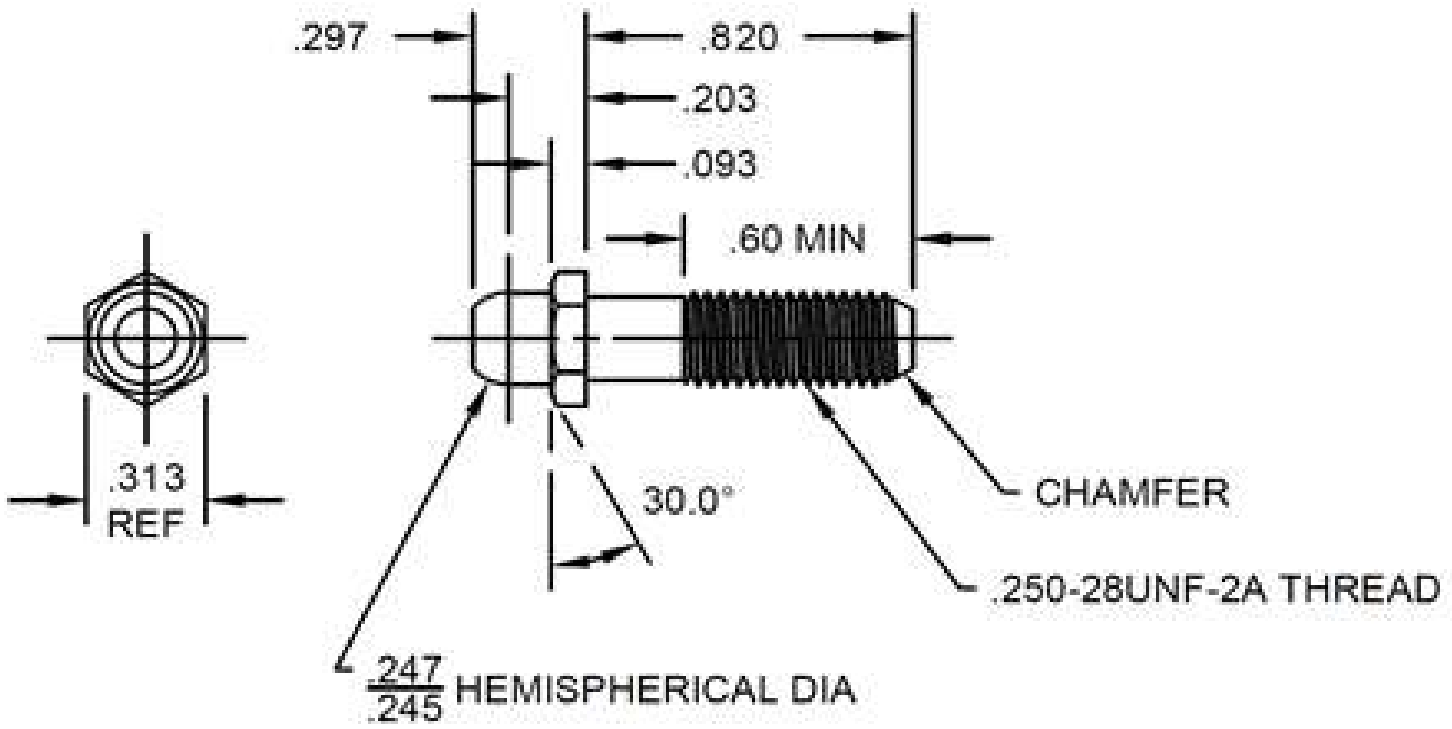


NOTES:

1. Areas in contact with the potting compound and gasket (figure 11) shall be sand blasted using 100 grit. Maximum removal of material shall be 0.001.
2. Apply enamel, commercial grade, to phase letters: A-Black, B-White, C-Red.

FIGURE 5. Insert plug. 04

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:33Z
Check the source to verify that this is the current version before use.



NOTES:

1. All fillets corner radii shall be a maximum of 0.005.
2. Finish shall be $\sqrt{63}$ on all machined surfaces except as shown.
3. Pin shall be tellurium copper, UNS C14500 in accordance with ASTM B301/B301M. Silver plate shall be in accordance with ASTM B700, type 2, grade B, class S, 0.0002 to 0.0003 thick. A nickel or nickel alloy intermediate layer, at least 0.00004 thick, shall be applied before the silver electroplate.

FIGURE 6. Contact pin. 05

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:33Z
Check the source to verify that this is the current version before use.

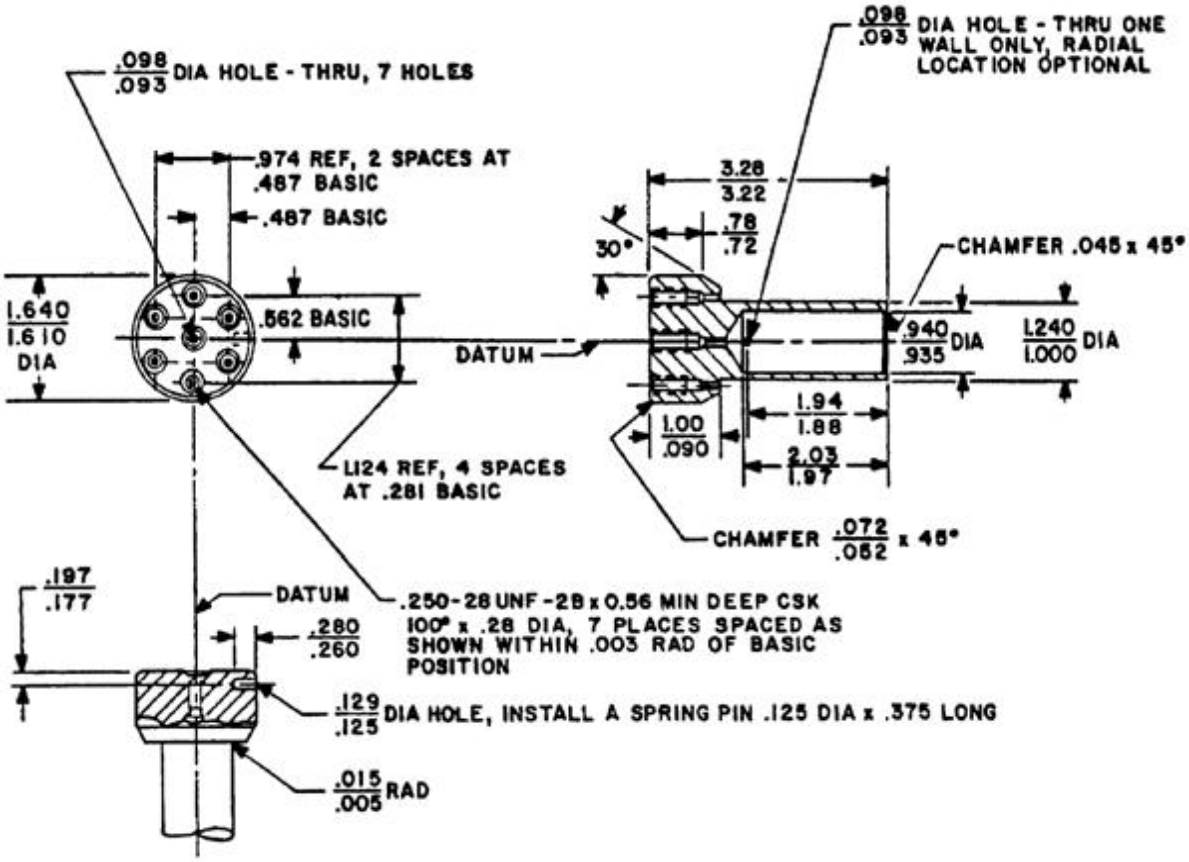


FIGURE 7. Bus lug. (06)

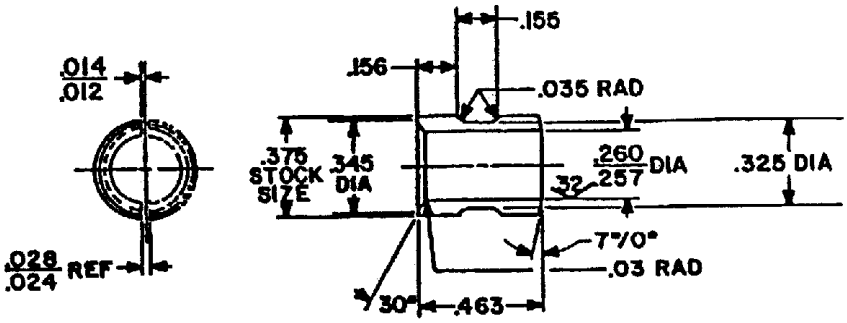
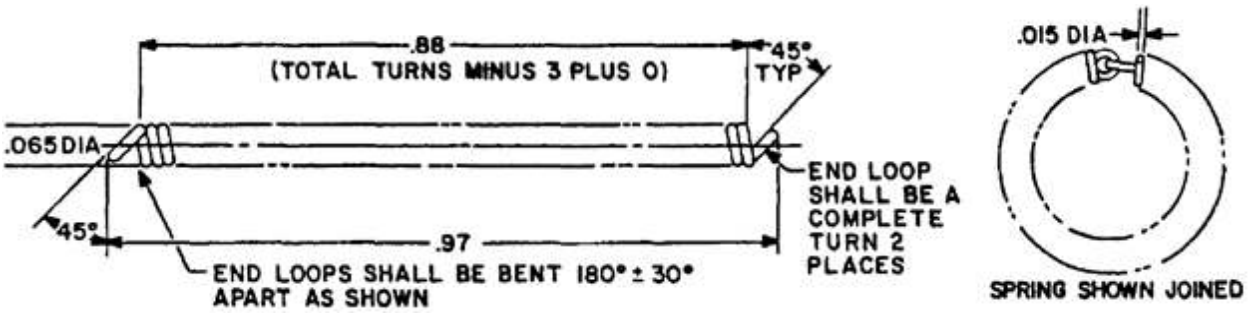


FIGURE 8. Contact sleeve segment. (07)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:33Z
Check the source to verify that this is the current version before use.

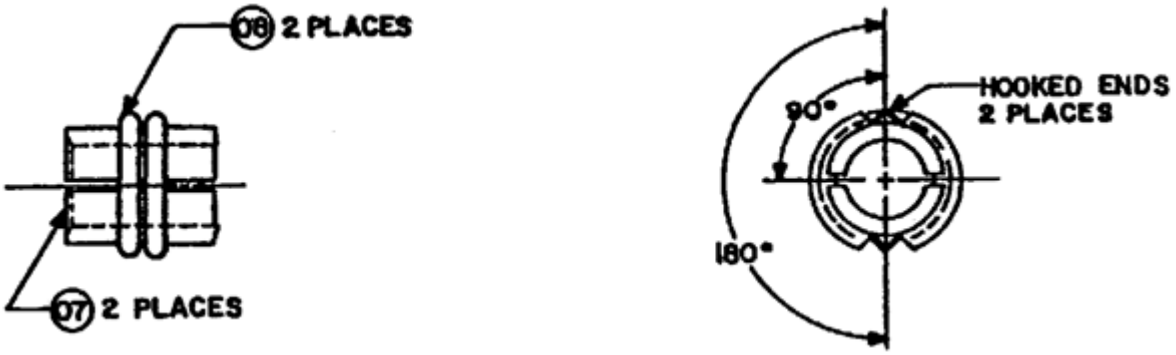


TABULATION

NOTES:

1. Total number of turns: 61
2. Coil mean diameter: 0.050
3. Spring rate (ref) pound/inch: 8.50
4. Arbor diameter: 0.375
5. Working length (ref): 1.23
6. Working load (ref) pound: 2.20

FIGURE 9. Spring. (08)



NOTES:

1. Spring hooks shall be circular in shape after assembly and show no visible evidence (bends, kinks, or twists) of an over-stressed condition. Hooks shall be securely engaged.
2. Hooked ends of springs shall be positioned 90 degrees from gap in segment and 180 degrees from each other as shown.

FIGURE 10. Contact sleeve assembly. (09)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:33Z
Check the source to verify that this is the current version before use.

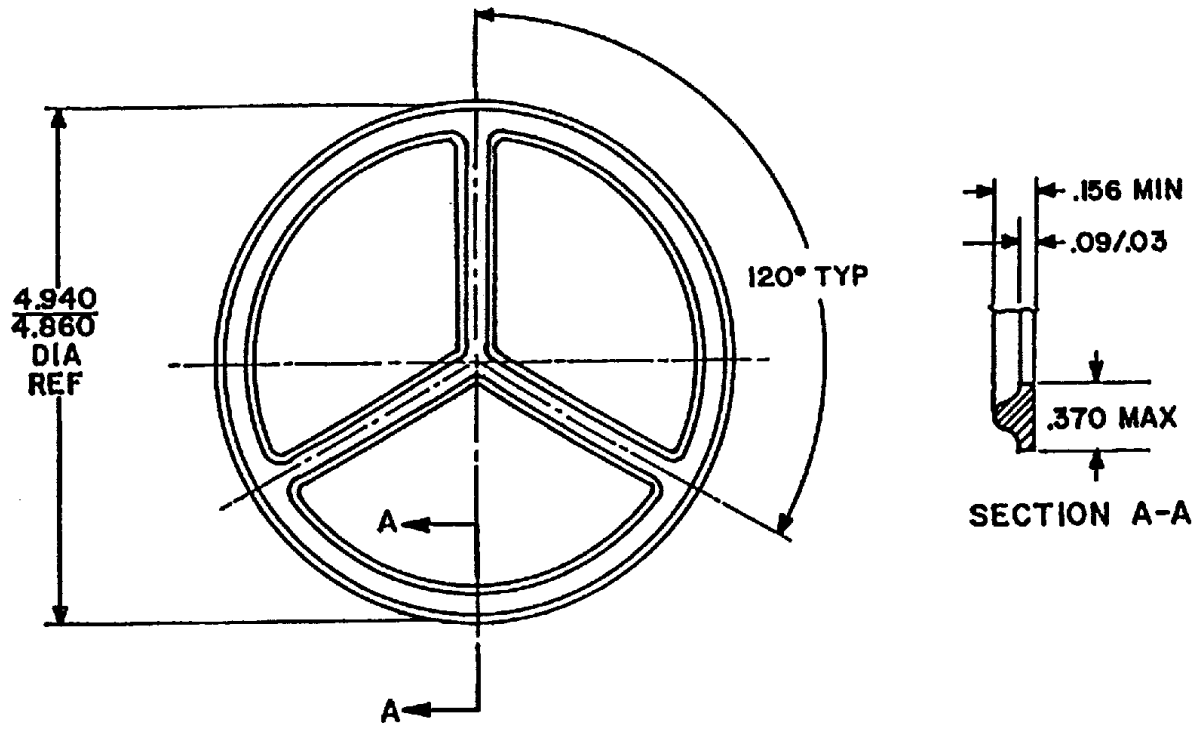


FIGURE 11. Gasket, insert. (10)

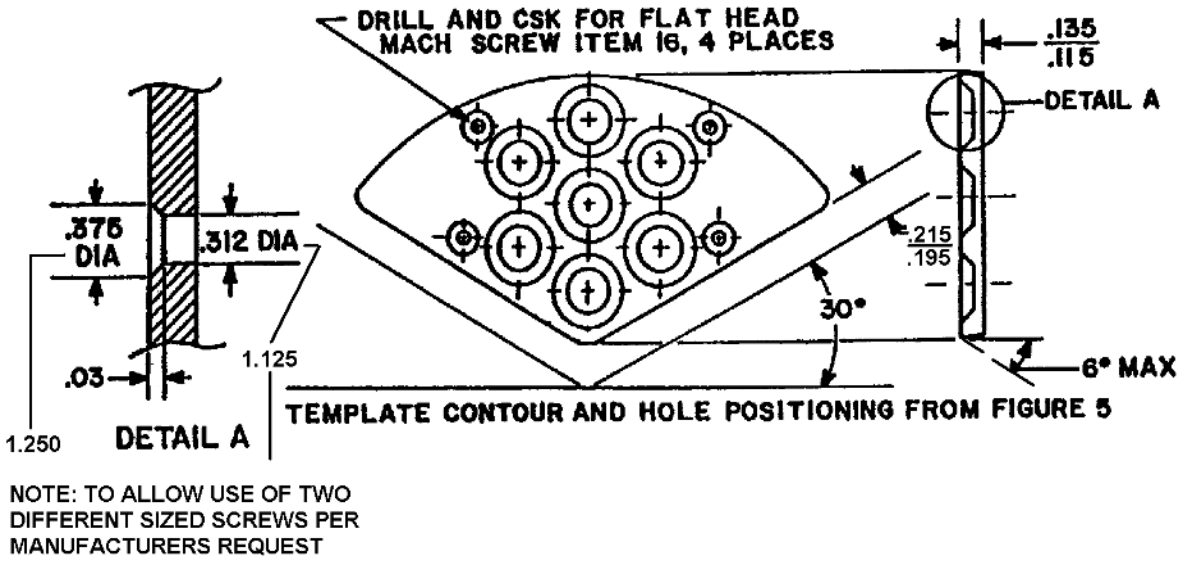


FIGURE 12. Sleeve retaining plate. (11)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:33Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

1. Electrical rating: 500 volts, 500 amperes, 60 hertz.

2. Dimensions and configuration: See figures 1 through 12.

3. Housing assembly: The plug housing and cover (see figures 2 or 3) shall be rubber in accordance with ASTM D2000. The color shall be black. Metallic reinforcing members shall be used. The housing shall provide a hinge and a mechanical latch for locking the cover to the plug or the plug to a receptacle (see MIL-DTL-24368/2). The final motion of the latch mechanism shall actuate the control switch or switches during lock-up.

4. Plug inserts: The insert and retaining plate material shall be as shown on figure 5. Inserts shall have provision for sealing between phases and the external environment when plug and receptacle (see MIL-DTL-24368/2) are mated or when cover is closed.

5. Contacts: Plug contacts shall be sockets and consist of sleeves that are split the entire length and retained together by springs around the periphery. The split sleeves shall be mounted to pins, the tips of which shall be finished to a hemispherical radius with an allowable flat of 0.123 inch less than the pin diameter. The pin shall be mechanically and rigidly held in the insert. The sleeve to pin mounting shall allow the contact a minimum radial side play of 0.020 inch at room temperature. The springs around the sleeves shall not be part of the electrical circuit. The contacts shall be in multiples per phase and terminate into a common bus bar phase (see figures 5 through 10).

6. Potting: Potting procedures shall be in accordance with MIL-STD-2003-3. When a plug is to be provided with the cable (see table II), the potting shall be Loctite Stycast 2741BK with Stycast 2741 W1 – Catalyst 15, or equal, or the cable may be molded to the plug at the manufacturer's choice.

7. Standard parts: The standard parts shown in table I shall be used in the construction of the plugs shown herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:33Z
Check the source to verify that this is the current version before use.

TABLE I. Standard parts list.

Item no. 1/	Description	Quantity	Material/Specification	Remarks
01	Plug	---	--	figure 1
02	Plug housing assembly with cover	1	See construction requirement	figure 2
03	45-degree plug housing assembly with cover	1	See construction requirement	figure 3
04	Insert, plug	1	3/	figure 5
05	Pin, contact, electric	21	Copper/ASTM B301/B301M	figure 6
06	Bus lug, 500MCM 2/	3	Copper/ASTM B301/B301M 2/	figure 7
07	Contact sleeve segment	42	Copper/ASTM B301/B301M	figure 8
08	Spring	42	ASTM A313/A313M	figure 9
09	Contact sleeve assembly	21	---	figure 10
10	Gasket insert	1	---	figure 11
11	Plate, sleeve retaining	3	3/	figure 12
12	Screw	8	NASM16996-16	---
13	Washer, lock	8	NASM35338-138	---
14	Washer, flat	8	NAS620C10L	---
15	Washer, lock	21	NASM35338-139	---
16	Washer, flat	21	NAS620C416L	---
17	Screw	12	NASM24693-C25	---
18	Guide pin	1	NASM16556-627	---
19	Cable, LSTHOF-500 4/	10 feet	MIL-DTL-24643/3	---
20	Potting	1 gallon	See potting requirement	---
21	Band	1	CRES, commercial	---
22	Insert, screw	12	CRES (for item 17)	---
23	Instruction plate	1	5/	---

NOTES:

1/ Item numbers are the same as numbers shown in circles on figures.

2/ Alloy number C14500 or C18700 shall be used.

3/ Molding material shall be glass roving reinforced phenolic, color black, Sumitomo GF 8202 L24, or equal.

4/ For PINs M24368/1–003 and M24368/1–005.

5/ Instruction plate shall be CRES with adhesive back and located on the band item 21.

8. Cable: Unless otherwise specified in the acquisition document, cable length shall be 10 feet ±6 inches as measured from the front of the plug to the end of the cable.

9. Impact: In addition to the requirements of MIL-C-24368, the handle shall not release.

10. High-impact shock: N/A.

11. Mechanical vibration: N/A.

12. Physical shock: In addition to the requirements of MIL-C-24368, the handle shall not release.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:33Z
Check the source to verify that this is the current version before use.

GENERAL INFORMATION:

1. Part or Identifying Numbers (PINs) are created as shown in table II.

TABLE II. Assignment of PINs.

PIN	Item name
M24368/1–001	Discontinued
M24368/1–002	Connector, plug, shore power
M24368/1–003	Connector, plug, shore power, with 10 feet of LSTHOF-500 cable
M24368/1–004	45-degree connector, plug, shore power
M24368/1–005	45-degree connector, plug, shore power, with 10 feet of LSTHOF-500 cable

2. Typical installation of shore power plugs is shown in MIL-STD-2003-2 and MIL-STD-2003-3.

3. Contact information for the part numbers specified herein is as follows:

Henkel Corp. (CAGE 80UG0) https://www.henkel.com (908) 685-7000 10 Finderne Avenue Bridgewater, NJ 08807	Sumitomo Bakelite North America, Inc. (CAGE 1JX75) https://www.sbhpp.com (248) 313-7000 46820 Magellan Drive, Suite C Novi, MI 48377

ACQUISITION REQUIREMENTS:

1. Length shall be specified when more than 10 feet of cable is required.

CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Navy – SH

Review activity:
Navy – EC

Preparing activity:
Navy – SH
(Project 6150-2020-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:33Z
Check the source to verify that this is the current version before use.